

1.) Algorithm to Convert PDA to CFG.

1) Input: PDA $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$ where:

$Q \rightarrow$ set of states

$\Sigma \rightarrow$ input Alphabet

$\Gamma \rightarrow$ stack Alphabet

$\delta \rightarrow$ Transition function

$q_0 \rightarrow$ start state

$z_0 \rightarrow$ initial stack symbol

$F \rightarrow$ set of Accepting states.

2) Output: CFG $G = (V, T, P, S)$ where

$V \rightarrow$ set of variables

$T \rightarrow$ Terminal Alphabet (Σ)

$P \rightarrow$ set of productions

$S \rightarrow$ start symbol.

3) Steps:

1) Define the terminal set: $T = \Sigma$

2) Define the variable set: $[p, A, q]$ where:

$p, q \in Q$

$A \in T$

$[p, A, q]$ represents the language of strings that take PDA from state p with A on top of the stack to state q with A popped.

3) Define the start symbol: $S = [q_0, z_0, q] \forall q \in Q$. PDA starts with q_0 with z_0

4) Construct productions:

$\rightarrow$ For each transition $\delta(p, A, \gamma) = (r, B_1 B_2 \dots B_k)$ (where $e \in \Sigma \cup \{\epsilon\}$, $k \geq 0$, $B_i \in \Gamma$).

$k=0$ (popping A with no push), add $[p, A, q] \rightarrow \alpha$.

$k=1$ (replace A with B_1), add $[p, A, q] \rightarrow \alpha [1, B_1, q] \forall q \in Q$.

$k=2$ (push $B_1 B_2$), add $[p, A, q] \rightarrow \alpha [r, B_1, q] [s, B_2, q] \forall s, r \in Q$.

Generalize $k > 2$ similarly, splitting into k variables.

→ Simplify: Remove unreachable and productions that cannot lead to terminals

4) Result: The CFG generates the same language as the PDA accepts by empty stack.

2.) Apply the above algorithm to convert PDA P to CFG G :

Given PDA: $P = (\{X, Y, Z, \epsilon\}, \{0, 1\}, \{A, Z\}, \delta, X, Z, \emptyset)$. with transitions:

$$\delta(X, 0, Z) = (X, AZ)$$

$$\delta(X, 0, A) = (X, AA)$$

$$\delta(X, 1, A) = (Y, \epsilon)$$

$$\delta(Y, 1, A) = (Y, \epsilon)$$

$$\delta(Y, \epsilon, A) = (Y, \epsilon)$$

$$\delta(Y, \epsilon, Z) = (Y, \epsilon)$$

This PDA accepts by empty stack ($F = \emptyset$)

→ Terminals $T = \{0, 1\}$

→ Variables: $V = \{[q, B, q] \mid q, q' \in \{X, Y\}, B \in \{A, Z\}\} \cup \{[X, A, X], [X, A, Y], [X, Z, X], [X, Z, Y], [Y, A, X], [Y, A, Y], [Y, Z, X], [Y, Z, Y]\}$

→ Start symbol: S generates $[X, Z, X]$ & $[X, Z, Y]$

→ Construct:

$$S \rightarrow [X, Z, X]$$

$$S \rightarrow [X, Z, Y]$$

For $\delta(X, 0, Z) = (X, AZ)$ (push A then $z_1 k = 2$):

$$[X, Z, \epsilon] \rightarrow 0 [X, A, \epsilon] [Z, Z, \epsilon] \quad \forall \epsilon \in \{X, Y\}$$

$$\rightarrow [X, Z, X] \rightarrow 0 [X, A, X] [X, Z, X]$$

$$\rightarrow [X, Z, X] \rightarrow 0 [X, A, Y] [Y, Z, X]$$

$$\rightarrow [X, Z, Y] \rightarrow 0 [X, A, X] [Y, Z, Y]$$

$$\rightarrow [Y, Z, Y] \rightarrow 0 [X, A, Y] [Y, Z, Y]$$

For $\delta(X, 0, A) = (X, AA)$ (push A then $A_1 k = 2$):

$$\therefore [X, A, \epsilon] \rightarrow 0 [X, A, \epsilon] [Z, A, \epsilon] \quad \forall \epsilon \in \{X, Y\}$$

$$\rightarrow [X, A, X] \rightarrow 0 [X, A, X] [X, A, X]$$

$$\rightarrow [X, A, X] \rightarrow 0 [X, A, Y] [Y, A, X]$$

$$\rightarrow [X, A, Y] \rightarrow 0 [X, A, X] [X, A, Y]$$

$$\rightarrow [x_1 A M] \rightarrow 0 [x_1 A M] [y_1 A M]$$

For $\delta(x_1, A) = (y_1, \varepsilon) \text{ (pop } A | k=0)$:

$$\rightarrow [x_1 A M] \rightarrow 1$$

For $\delta(y_1, A) = (y_1, \varepsilon) \text{ (pop } A | k=0)$:

$$\rightarrow [y_1 A M] \rightarrow 1$$

For $\delta(y_1, \varepsilon, A) = (y_1, \varepsilon) \text{ (pop } A | k=0)$:

$$\rightarrow [y_1 A M] \rightarrow \varepsilon$$

For $\delta(y_1, \varepsilon, z) = (y_1, \varepsilon) \text{ (pop } z | k=0)$:

$$\rightarrow [y_1 z M] \rightarrow \varepsilon$$

simplify

$$S \rightarrow [x_1 z X] \text{ \& } S \rightarrow [x_1 z M]$$

from $[x_1 z X]$:

$$\rightarrow [A z X] \rightarrow 0 [x_1 A X] [x_1 z X]$$

$$\rightarrow [x_1 z X] \rightarrow 0 [x_1 A M] [y_1 z X] \rightarrow ([x_1 A M], [y_1 z X])$$

from $[x_1 z Y]$:

$$\rightarrow [x_1 z X] \rightarrow 0 [x_1 A X] [x_1 z Y] \text{ (uses } [x_1 A X])$$

$$\rightarrow [x_1 z M] \rightarrow 0 [x_1 A Y] [y_1 z Y] \text{ (uses } [x_1 A M], [y_1 z Y])$$

from $[x_1 A M]$:

$$\rightarrow [x_1 A M] \rightarrow 1$$

$$\rightarrow [x_1 A M] \rightarrow 0 [x_1 A X] [x_1 A M] \text{ (uses } [x_1 A X])$$

$$\rightarrow [x_1 A M] \rightarrow 0 [x_1 A Y] [y_1 A M] \text{ (uses } [y_1 A M])$$

from $[y_1 z Y]$:

$$\rightarrow [y_1 z M] \rightarrow \varepsilon$$

from $[y_1 A M]$:

$$\rightarrow [y_1 A M] \rightarrow 1$$

$$\rightarrow [y_1 A M] \rightarrow \varepsilon$$

$$[x_1 A X], [y_1 A X], [y_1 z X]$$

↳ don't lead to terminate without looping infinitely or not reachable from S.

→ Final CFG G :

1) $S \rightarrow [X_1 Z_1 Y]$

2) $[X_1 Z_1 Y] \rightarrow 0 [X_1 A_1 Y] [Y_1 Z_1 Y]$

3) $[X_1 A_1 Y] \rightarrow 1$

4) $[X_1 A_1 Y] \rightarrow 0 [X_1 A_1 Y] [Y_1 A_1 Y]$

5) $[Y_1 A_1 Y] \rightarrow 1$

6) $[Y_1 A_1 Y] \rightarrow \epsilon$

7) $[Y_1 Z_1 Y] \rightarrow \epsilon$

rename:

$S \rightarrow S(\text{start})$

$[X_1 Z_1 Y] \rightarrow A,$

$[X_1 A_1 Y] \rightarrow B,$

$[Y_1 A_1 Y] \rightarrow C,$

$[Y_1 Z_1 Y] \rightarrow D.$

productions:

1) $S \rightarrow A$

2) $A \rightarrow 0BD$

3) $B \rightarrow 1$

4) $B \rightarrow 0BC$

5) $C \rightarrow 1$

6) $C \rightarrow \epsilon$

7) $D \rightarrow \epsilon$

Final CFG:

$G = (V, \Gamma, P, S)$ where:

$V = \{S, A, B, C, D\}$

$\Gamma = \{0, 1, \epsilon\}$

$P =$

$S \rightarrow A$

$A \rightarrow 0BD$

$B \rightarrow 1 \mid 0BC$

$C \rightarrow 1 \mid \epsilon$

$D \rightarrow \epsilon$

$S \rightarrow$ Start symbol

This CFG generates the same language as PDA which accepts strings like $0^n 1^n$, pushing A s for 0 s and popping them for 1 s.